

INFS3200 A2

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1. Task 1

2. Q1

```
from a2.src import P4
from a2.src.helpers import nested_loop
from p4.DataLinkage_py.src.data.db_loader import db_loader
from p4.DataLinkage_py.src.data.measurement import calc_measure, load_benchmark

# REF: https://en.wikibooks.org/wiki/Algorithm_Implementation/Strings/Levenshtein_distance#Python
def levenshtein(s1, s2):
    if len(s1) < len(s2):
        return levenshtein(s2, s1)

    # len(s1) >= len(s2)
    if len(s2) == 0:
        return len(s1)

    previous_row = range(len(s2) + 1)
    for i, c1 in enumerate(s1):
        current_row = [i + 1]
        for j, c2 in enumerate(s2):
            insertions = (
                previous_row[j + 1] + 1
            ) # j+1 instead of j since previous_row and current_row are one character longer
            deletions = current_row[j] + 1 # than s2
            substitutions = previous_row[j] + (c1 != c2)
            current_row.append(min(insertions, deletions, substitutions))
        previous_row = current_row

    return previous_row[-1]

if __name__ == "__main__":
    restaurants = db_loader()
    benchmark = load_benchmark(P4 / "data" / "restaurant_pair.csv")
    for t in [0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]:
        count = 0
        res = []
        for i, j in nested_loop(restaurants):
            name1 = i.get_name()
            name2 = j.get_name()
            edi_dist = levenshtein(name1, name2)
            sim = 1 - (edi_dist / max(len(name1), len(name2)))
            if sim >= t:
                count += 1
                res.append(f"{str(i.get_id())}_{str(j.get_id())}")

    print(f"Threshold: {t} similar count: {count}")
    calc_measure(res, benchmark)
    print("-----")
```

```

s4696303@infs3200-0b490d8e:~/infs3200$ uv run python -m a2.src.task1.q1
PostgreSQL database version:
('PostgreSQL 17.7 (Ubuntu 17.7-3.pgdg24.04+1) on x86_64-pc-linux-gnu, compiled by gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0, 64-bit',)
Threshold: 0.4 similar count: 4142
Precision= 0.022935779816513763 , Recall= 0.8962264150943396 , Fmeasure= 0.04472693032015067
-----
Threshold: 0.5 similar count: 1349
Precision= 0.06523350630096368 , Recall= 0.8301886792452831 , Fmeasure= 0.12096219931271479
-----
Threshold: 0.6 similar count: 359
Precision= 0.23119777158774374 , Recall= 0.7830188679245284 , Fmeasure= 0.35698924731182796
-----
Threshold: 0.7 similar count: 139
Precision= 0.5683453237410072 , Recall= 0.7452830188679245 , Fmeasure= 0.6448979591836734
-----
Threshold: 0.8 similar count: 97
Precision= 0.8144329896907216 , Recall= 0.7452830188679245 , Fmeasure= 0.7783251231527093
-----
Threshold: 0.9 similar count: 86
Precision= 0.8953488372093024 , Recall= 0.7264150943396226 , Fmeasure= 0.8020833333333334
-----
Threshold: 1.0 similar count: 82
Precision= 0.9146341463414634 , Recall= 0.7075471698113207 , Fmeasure= 0.7978723404255319
-----
s4696303@infs3200-0b490d8e:~/infs3200$ 

```

2.1. Q2

```

from a2.src import P4
from a2.src.helpers import nested_loop
from p4.DataLinkage_py.src.data.db_loader import db_loader
from p4.DataLinkage_py.src.data.measurement import calc_measure, load_benchmark

def qgram(str: str, q: int) -> set[str]:
    return {str[i : i + 1] for i in range(len(str) - q + 1)}

def jaccard(qgram1: set, qgram2: set):
    if not qgram1 or not qgram2:
        return 1.0
    return len(qgram1 & qgram2) / len(qgram1 | qgram2)

if __name__ == "__main__":
    restaurants = db_loader()
    benchmark = load_benchmark(P4 / "data" / "restaurant_pair.csv")
    for q in [2, 3, 4]:
        for t in [0.6, 0.7, 0.8, 0.9, 1.0]:
            count = 0
            res = []
            for i, j in nested_loop(restaurants):
                name1 = i.get_name()
                name2 = j.get_name()
                sim = jaccard(qgram(name1, q), qgram(name2, q))
                if sim >= t:
                    count += 1
                    res.append(f"{str(i.get_id())}_{str(j.get_id())}")

            print(f"Q: {q} Threshold: {t} similar count: {count}")
            calc_measure(res, benchmark)
            print("-----")

```

```

s4696303@infs3200-0b490d8e:~/infs3200$ uv run python -m a2.src.task1.q2
PostgreSQL database version:
('PostgreSQL 17.7 (Ubuntu 17.7-3.pgdg24.04+1) on x86_64-pc-linux-gnu, compiled by gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 1
3.3.0, 64-bit',)
Q: 2 Threshold: 0.6 similar count: 7172
Precision= 0.01324595649749024 , Recall= 0.8962264150943396 , Fmeasure= 0.026106073097004672
-----
Q: 2 Threshold: 0.7 similar count: 1270
Precision= 0.07086614173228346 , Recall= 0.8490566037735849 , Fmeasure= 0.1308139534883721
-----
Q: 2 Threshold: 0.8 similar count: 259
Precision= 0.3281853281853282 , Recall= 0.8018867924528302 , Fmeasure= 0.4657534246575343
-----
Q: 2 Threshold: 0.9 similar count: 103
Precision= 0.7864077669902912 , Recall= 0.7641509433962265 , Fmeasure= 0.7751196172248804
-----
Q: 2 Threshold: 1.0 similar count: 92
Precision= 0.8586956521739131 , Recall= 0.7452830188679245 , Fmeasure= 0.7979797979797979
-----
Q: 3 Threshold: 0.6 similar count: 7438
Precision= 0.012368916375369724 , Recall= 0.8679245283018868 , Fmeasure= 0.024390243902439025
-----
Q: 3 Threshold: 0.7 similar count: 2737
Precision= 0.03251735476799415 , Recall= 0.839622641509434 , Fmeasure= 0.06260991909954274
-----
Q: 3 Threshold: 0.8 similar count: 1943
Precision= 0.04323211528564076 , Recall= 0.7924528301886793 , Fmeasure= 0.08199121522693997
-----
Q: 3 Threshold: 0.9 similar count: 1820
Precision= 0.04450549450549451 , Recall= 0.7641509433962265 , Fmeasure= 0.08411214953271028
-----
Q: 3 Threshold: 1.0 similar count: 1811
Precision= 0.04251794588625069 , Recall= 0.7264150943396226 , Fmeasure= 0.08033385498174231
-----
Q: 4 Threshold: 0.6 similar count: 9912
Precision= 0.009180790960451978 , Recall= 0.8584905660377359 , Fmeasure= 0.018167298862048313
-----
Q: 4 Threshold: 0.7 similar count: 6047
Precision= 0.0143872994873491 , Recall= 0.8207547169811321 , Fmeasure= 0.028278888347147736
-----
Q: 4 Threshold: 0.8 similar count: 5391
Precision= 0.015396030421072158 , Recall= 0.7830188679245284 , Fmeasure= 0.030198289976350736
-----
Q: 4 Threshold: 0.9 similar count: 5268
Precision= 0.015186028853454821 , Recall= 0.7547169811320755 , Fmeasure= 0.0297729810197246
-----
Q: 4 Threshold: 1.0 similar count: 5263
Precision= 0.01444043321299639 , Recall= 0.7169811320754716 , Fmeasure= 0.02831067237846899
-----

```

2.2. Q3

```

from a2.src import P4
from a2.src.helpers import nested_loop
from a2.src.task1.q2 import jaccard, qgram
from p4.DataLinkage_py.src.data.db_loader import db_loader
from p4.DataLinkage_py.src.data.measurement import calc_measure, load_benchmark

```

```

if __name__ == "__main__":
    restaurants = db_loader()
    benchmark = load_benchmark(P4 / "data" / "restaurant_pair.csv")
    q = 3
    max_f_measure = 0
    max_a = 0
    for a in [0.5, 0.7, 0.9]:
        for t in [0.5, 0.7, 0.9]:
            count = 0
            res = []
            for i, j in nested_loop(restaurants):
                name1 = i.get_name()

```

```

addr1 = i.get_address()
name2 = j.get_name()
addr2 = j.get_address()
sim_name = jaccard(qgram(name1, q), qgram(name2, q))
sim_addr = jaccard(qgram(addr1, q), qgram(addr2, q))
sim = a * sim_name + (1 - a) * sim_addr
if sim >= t:
    count += 1
    res.append(f"{str(i.get_id())}_{str(j.get_id())}")

print(f"Q: {q} Threshold: {t} similar count: {count}")
_, _, f_measure = calc_measure(res, benchmark)
print("-----")
if f_measure > max_f_measure:
    max_f_measure = f_measure
    max_a = a
print(f"Max pair ({max_a}, {max_f_measure})")

```

```

s4696303@infs3200-0b490d8e:~/infs3200$ uv run python -m a2.src.task1.q3
PostgreSQL database version:
('PostgreSQL 17.7 (Ubuntu 17.7-3.pgdg24.04+1) on x86_64-pc-linux-gnu, compiled by gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0, 64-bit',)
Q: 3 Threshold: 0.5 similar count: 8667
Precision= 0.011999538479289258 , Recall= 0.9811320754716981 , Fmeasure= 0.02370910748888636
-----
Q: 3 Threshold: 0.7 similar count: 1208
Precision= 0.0802980132450331 , Recall= 0.9150943396226415 , Fmeasure= 0.1476407914764079
-----
Q: 3 Threshold: 0.9 similar count: 64
Precision= 0.90625 , Recall= 0.5471698113207547 , Fmeasure= 0.6823529411764705
-----
Q: 3 Threshold: 0.5 similar count: 11012
Precision= 0.009081002542680712 , Recall= 0.9433962264150944 , Fmeasure= 0.017988846914912753
-----
Q: 3 Threshold: 0.7 similar count: 1934
Precision= 0.047569803516028956 , Recall= 0.8679245283018868 , Fmeasure= 0.09019607843137256
-----
Q: 3 Threshold: 0.9 similar count: 71
Precision= 0.8732394366197183 , Recall= 0.5849056603773585 , Fmeasure= 0.7005649717514125
-----
Q: 3 Threshold: 0.5 similar count: 18353
Precision= 0.005339726475235656 , Recall= 0.9245283018867925 , Fmeasure= 0.010618126659082289
-----
Q: 3 Threshold: 0.7 similar count: 2224
Precision= 0.04091726618705036 , Recall= 0.8584905660377359 , Fmeasure= 0.07811158798283262
-----
Q: 3 Threshold: 0.9 similar count: 1815
Precision= 0.04462809917355372 , Recall= 0.7641509433962265 , Fmeasure= 0.08433107756376887
-----
Max pair (0.7, 0.7005649717514125)
s4696303@infs3200-0b490d8e:~/infs3200$

```